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5. Relationship Between Events

Since:

P(Pass | Group) ≠ P(Pass)

So, events are Dependent ¶

Also, they are not mutually exclusive

6. Final Insights

Students who study more than 10 hours are more likely to pass

Attendance above 80% increases success

Group discussion improves performance

Students with all three factors have highest success rate

In [7]:
```
df = pd.read_csv(r"C:\Users\tanaa\Desktop\students_dataset_200.csv")
df
```

Out[7]:

	study_hours	attendance	group_discussion	previous_test_score	final_exam_pass
0	10	81	Yes	57	Fail
1	7	67	Yes	41	Fail
2	16	95	No	91	Pass
3	14	57	Yes	67	Fail
4	11	93	Yes	69	Pass

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In [7]:
```
df = pd.read_csv(r"C:\Users\tanaa\Desktop\students_dataset_200.csv")
df
```

Out[7]:

	study_hours	attendance	group_discussion	previous_test_score	final_exam_pass
0	10	81	Yes	57	Fail
1	7	67	Yes	41	Fail
2	16	95	No	91	Pass
3	14	57	Yes	67	Fail
4	11	93	Yes	69	Pass
...	...	...	...	...	...
195	15	69	No	71	Fail
196	12	82	Yes	69	Pass
197	10	88	Yes	62	Fail
198	4	56	Yes	30	Fail
199	4	86	No	33	Fail

200 rows × 5 columns

In [8]:
```
df.head()
```

Out[8]:

	study_hours	attendance	group_discussion	previous_test_score	final_exam_pass
0	10	81	Yes	57	Fail
1	7	67	Yes	41	Fail
2	16	95	No	91	Pass
3	14	57	Yes	67	Fail
4	11	93	Yes	69	Pass

In [9]:
```
df.tail()
```

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In [9]:
```
df.tail()
```

Out[9]:

	study_hours	attendance	group_discussion	previous_test_score	final_exam_pass
195	15	69	No	71	Fail
196	12	82	Yes	69	Pass
197	10	88	Yes	62	Fail
198	4	56	Yes	30	Fail
199	4	86	No	33	Fail

In [10]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   study_hours          200 non-null    int64
 1   attendance           200 non-null    int64
 2   group_discussion     200 non-null    str
 3   previous_test_score  200 non-null    int64
 4   final_exam_pass      200 non-null    str
dtypes: int64(3), str(2)
memory usage: 7.9 KB
```

In [11]:
```
df.dtypes
```

Out[11]:
```
study_hours            int64
attendance             int64
group_discussion         str
previous_test_score    int64
final_exam_pass          str
dtype: object
```

In [12]:
```
pass_prob = df[df["final_exam_pass"] == "Pass"]
pass_prob
```

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In [12]:
```
pass_prob = df[df["final_exam_pass"] == "Pass"]
pass_prob
```

Out[12]:

	study_hours	attendance	group_discussion	previous_test_score	final_exam_pass
2	16	95	No	91	Pass
4	11	93	Yes	69	Pass
11	14	78	No	85	Pass
13	11	86	Yes	63	Pass
16	11	95	No	74	Pass
26	15	77	No	76	Pass
29	15	87	Yes	94	Pass
33	15	80	No	87	Pass
34	12	86	No	82	Pass
36	14	86	Yes	85	Pass
40	15	92	No	88	Pass
45	12	76	No	72	Pass
55	12	90	No	76	Pass
64	15	87	Yes	83	Pass
65	15	94	Yes	87	Pass
68	16	94	Yes	83	Pass
74	11	85	No	74	Pass
79	13	80	Yes	72	Pass
84	13	91	Yes	87	Pass
85	15	86	Yes	86	Pass
87	13	95	Yes	79	Pass
93	11	78	Yes	74	Pass
95	16	78	No	84	Pass
98	11	90	No	61	Pass

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In [14]:
```
pass_prob = (df["final_exam_pass"] == "Pass").var()
pass_prob
```

Out[14]:
```
np.float64(0.20902010050251263)
```

In [15]:
```
attendence_prob = df[df["attendance"] > 80]
attendence_prob
```

Out[15]:

	study_hours	attendance	group_discussion	previous_test_score	final_exam_pass
0	10	81	Yes	57	Fail
2	16	95	No	91	Pass
4	11	93	Yes	69	Pass
7	10	81	No	56	Fail
9	6	91	No	44	Fail
...	...	...	...	...	...
182	13	89	No	86	Pass
187	15	90	No	77	Pass
196	12	82	Yes	69	Pass
197	10	88	Yes	62	Fail
199	4	86	No	33	Fail

85 rows × 5 columns

In [16]:
```
attendence_prob = df["attendance"].mean()
attendence_prob
```

Out[16]:
```
np.float64(76.23)
```

In [17]:
```
attendence_prob = df["attendance"].var()
attendence_prob
```

Out[17]:
```
np.float64(147.1227135678392)
```

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In [18]:
```
study_prob = df[df["study_hours"] > 10]
study_prob
```

Out[18]:

	study_hours	attendance	group_discussion	previous_test_score	final_exam_pass
2	16	95	No	91	Pass
3	14	57	Yes	67	Fail
4	11	93	Yes	69	Pass
5	16	60	No	74	Fail
8	13	63	Yes	75	Fail
...	...	...	...	...	...
190	15	62	Yes	80	Fail
192	14	68	No	66	Fail
193	13	66	No	77	Fail
195	15	69	No	71	Fail
196	12	82	Yes	69	Pass

102 rows × 5 columns

In [19]:
```
study_prob = df['study_hours'].mean()
study_prob
```

Out[19]:
```
np.float64(10.46)
```

In [20]:
```
study_prob = df['study_hours'].var()
study_prob
```

Out[20]:
```
np.float64(13.96824120603015)
```

In [21]:
```
group_discussion_prob = df[df["group_discussion"] == "Yes"]
group_discussion_prob
```

Out[21]:

	study_hours	attendance	group_discussion	previous_test_score	final_exam_pass

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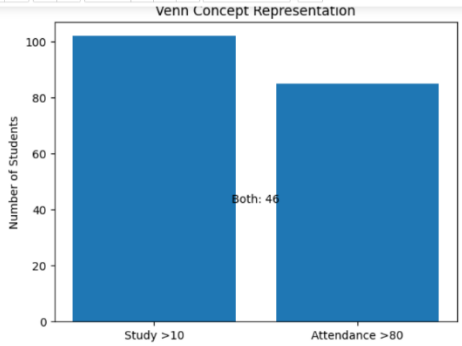

In [28]:
```
table = pd.crosstab(df['group_discussion'], df['final_exam_pass'])
table
```

Out[28]:

final_exam_pass	Fail	Pass
group_discussion		
No	76	31
Yes	65	28

In [29]:
```
joint = table.loc['Yes', 'Pass'] / len(df)
print(joint)
```

```
0.14
```